Quiz

1. Which of the following BEST describes the relationship between AI, Machine Learning, and Deep

- A. AI is a subset of Machine Learning, which is a subset of Deep Learning.
- B. Machine Learning is a subset of AI, and Deep Learning is a subset of Machine Learning.
- C. Deep Learning is a subset of AI, and Machine Learning is a subset of Deep Learning.
- D. AI, Machine Learning, and Deep Learning are distinct and unrelated fields.

Answer: B

Explanation: Machine Learning is a subset of AI, as it is a method for achieving AI. Deep Learning is a subset of Machine Learn

2. What is the primary goal of Artificial Intelligence (AI)?

- A. To develop machines that can learn from data without explicit programming.
- B. To create neural networks that mimic the human brain.
- C. To simulate human intelligence using computers.
- D. To generate new content like text, audio, and video.

Answer: C

Explanation: AI aims to replicate human cognitive functions and problem-solving abilities through computational methods.

3. Which characteristic is most indicative of Machine Learning?

- A. The use of explicit programming to define specific rules.
- B. The ability of machines to learn from data without being explicitly programmed.
- C. The reliance on pre-defined algorithms without any learning component.
- D. The creation of new content such as text and images.

Answer: B

Explanation: Machine Learning allows systems to improve their performance on a task through experience (data), rather than di

4. What distinguishes Deep Learning from traditional Machine Learning?

- A. Deep Learning focuses on generating new content, while Machine Learning focuses on predictions.
- B. Deep Learning uses neural networks with multiple layers to mimic the human brain.
- C. Deep Learning requires explicit programming, while Machine Learning does not.
- D. Deep Learning is a broader field than Machine Learning.

Answer: B

Explanation: Deep Learning utilizes complex, multi-layered neural networks to analyze data, drawing inspiration from the structu

5. What are 'foundation models' in the context of Generative AI?

- A. Early AI systems based on expert knowledge.
- B. Large models, like large language models, that power Generative AI and enable the creation of new content.
- C. Simple algorithms used for basic pattern recognition.
- D. Neural networks with a single layer used in Deep Learning.

Answer: B

Explanation: Foundation models are the core engines of Generative AI, allowing for the generation of various forms of content b